

CHEMISTRY**Q1** How many of the following are paramagnetic in nature $O_2, O_2^+, O_2^-, NO_2, CO, NO, [Ni(NH_3)_6]^{2+}, K_3[Co(CN)_6], [NiCl_4]^{2-}$ **Ans.** 7**Sol.** $O_2, O_2^+, O_2^- \rightarrow$ Paramagnetic $NO_2, NO \rightarrow$ Paramagnetic $[Ni(NH_3)_6]^{2+} \rightarrow 3d^8 \rightarrow$ Paramagnetic $[NiCl_4]^{2-} \rightarrow 3d^8$ (WFL) $\rightarrow$ Paramagnetic $CO, K_3[Co(CN)_6] \rightarrow 3d^6$ (SFL) $\rightarrow$ Diamagnetic**Q2** Calculate the spin magnetic moment of Mn_2O_3 **Ans.** 4.8**Sol.** $Mn_2^{+3}O_3 \rightarrow 3d^4$

$$\mu = \sqrt{4(4+2)}$$

$$= \sqrt{24} = 4.8$$

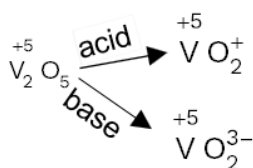
Q3 Consider the following oxides, V_2O_3, V_2O_4 , and V_2O_5 Change in oxidation state of vanadium when amphoteric oxide reacts with acids to form VO_4^{+} is

(1) 1

(2) 2

(3) 3

(4) 0

Ans. (4)**Sol.** V_2O_5 amphoteric**Q4** Which of the following is/are yellow and black in colour $CuS, PbS, As_2S_3, (NH_4)_2S$.(1) PbS, As_2S_3 (2) CuS, PbS (3) $As_2S_3, (NH_4)_2S$ (4) CuS, As_2S_3 **Ans.** (2) & (3)**Sol.** $CuS, PbS \rightarrow$ Black colour $As_2S_3, (NH_4)_2S \rightarrow$ Yellow colour

Q5 Which has maximum oxidizing power among the following

- (1) VO_2^+ (2) $\text{Cr}_2\text{O}_7^{2-}$ (3) MnO_4^- (4) TiO_2

Ans. (3)

Sol. $\overset{+5}{\text{V}}\text{O}_2^+$
 $\overset{+6}{\text{Cr}_2}\text{O}_7^{2-}$
 $\overset{+7}{\text{MnO}_4^-}$
 $\overset{+4}{\text{TiO}_2}$

Q6 Which of the following energy order is correct for H atom.

- (1) $1s < 2s < 2p < 3s$ (2) $1s < 2s = 2p < 3s = 3p$
 (3) $1s < 2s < 3s < 4s$ (4) $1s < 2s < 2p < 3p < 3s$

Ans. (2)

Sol. Energy is only decided by n.

Q7 Which of the group -15 element forms $d\pi - d\pi$ Bond and strongest basic hydride ?

- (1) $z = 7$ (2) $z = 15$ (3) $z = 33$ (4) $z = 51$

Ans. (2)

Sol. $\text{NH}_3 > \text{PH}_3 > \text{AsH}_3 > \text{SbH}_3$

Q8 How many of the following are polar

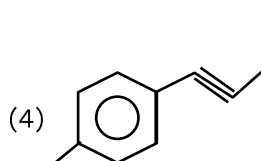
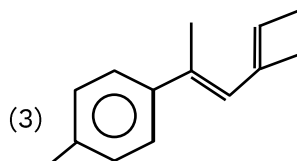
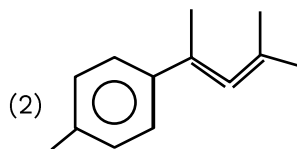
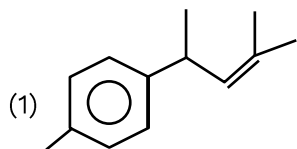
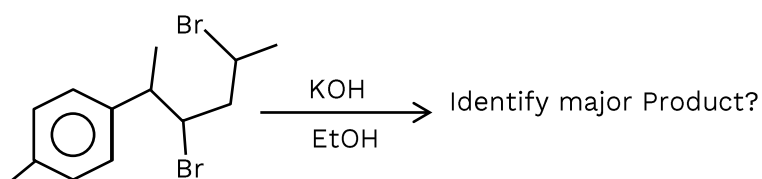
CH_4 , CCl_4 , CH_2Cl_2 , H_2O , NH_3 , H_2O_2 , O_2F_2

Ans. 5

Sol. CH_4 , $\text{CCl}_4 \rightarrow \mu = 0$ (non polar)

CH_2Cl_2 , H_2O , NH_3 , H_2O_2 , $\text{O}_2\text{F}_2 \rightarrow \mu \neq 0$ (Polar)

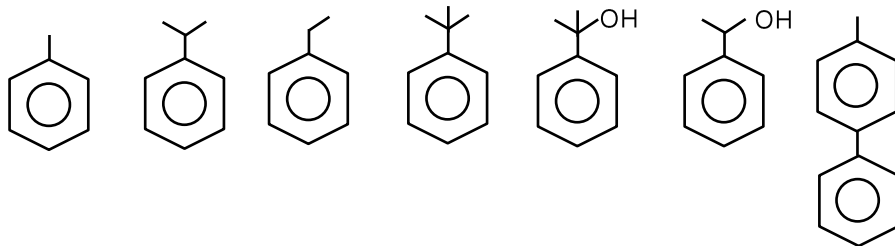
Q9



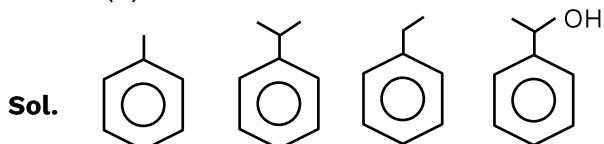
Ans. (3)

Sol. E_2 -elimination

Q10 Total number of compound react with KMnO_4 and give benzoic acid



Ans. (4)



Q11 The increasing solubility order of HgS , PbS , AgBr , Ca(OH)_2

- (1) $\text{HgS} < \text{PbS} < \text{AgBr} < \text{Ca(OH)}_2$ (2) $\text{PbS} < \text{HgS} < \text{Ca(OH)}_2 < \text{AgBr}$
 (3) $\text{AgBr} < \text{PbS} < \text{HgS} < \text{Ca(OH)}_2$ (4) $\text{Ca(OH)}_2 < \text{HgS} < \text{AgBr} < \text{PbS}$

Ans. (1)

Sol. Solubility order



Q12 30 gram HNO_3 is added to a solution to prepare 75% w/w solution having density 1.25 g/ml. Volume of solution is -

- (1) 48 ml (2) 36 ml (3) 28 ml (4) 32 ml

Ans. (4)

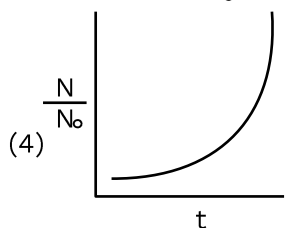
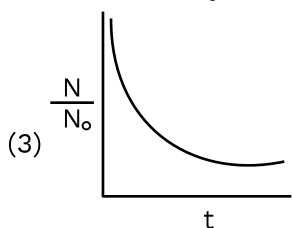
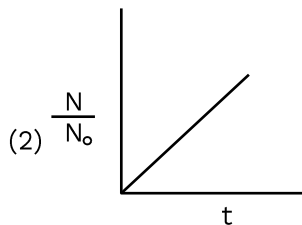
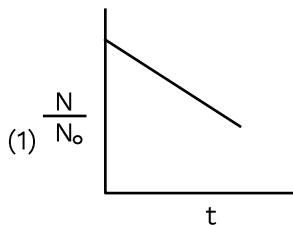
Sol. Final volume of sol. V_{m1}

mass of final sol = $V \times \text{density}$
 $v \times 1.25$

for 75% = $V \times 1.25 \times 0.75 = 30$

$v = \frac{30}{1.25} \times 0.75 = 32\text{ml}$

Q13 In a given bacteria decay, initial concentration is N and concentration at time t is N_0 . Find the graph between $\frac{N}{N_0}$ and t .



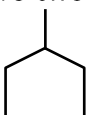
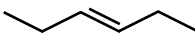
Ans. (3)**Sol.** According to 1st-order kinetics

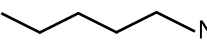
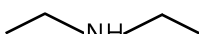
$$\lambda_t = \ln(N_0 / N)$$

$$= (N_0 / N) = e^{\lambda t}$$

$$= \left(\frac{N}{N_0} \right) = e^{-\lambda t}$$

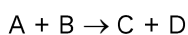
Q14 Given below are two statements :

statement (i):  &  These are isomer of each other

statement (ii):  &  are Functional isomer.

in the light of the above statements, choose the most appropriate answer from the options given below :

- (1) both statement i and statement ii are incorrect
- (2) statement i is correct but statement ii is incorrect
- (3) both statement i and statement ii are correct
- (4) statement i is incorrect but statement ii is correct

Ans. (3)**Sol.** Conceptual**Q15** For an elementary reaction

When volume becomes $\frac{1}{3}$ rd rate of reaction becomes

- (1) 6 times
- (2) 8 times
- (3) 9 times
- (4) 2 times

Ans. (3)**Sol.** $r = k[A][B]$ If $V \rightarrow V / 3$ than $C \rightarrow 3C$ So $r' = 9r$ **Q16** **List-I**

- (P) Maltose
- (Q) Lactose
- (R) Amylopectin
- (S) Sucrose

List-II

- (1) $C_1 \beta - C_4 \beta$
- (2) $C_1 \alpha - C_6 \alpha$
- (3) $C_1 \alpha - C_4 \alpha$
- (4) $C_1 \alpha - C_2 \beta$

- (A) P→3; Q→1; R→2; S→4
- (B) P→1; Q→3; R→2; S→4
- (C) P→3; Q→2; R→1; S→4
- (D) P→3; Q→1; R→4; S→2

Ans. (1)

Sol. Theory based

Q17 Match the following List-I with List-II

List-I

(P) $[\text{CoF}_6]^{3-}$

(Q) $[\text{Co}(\text{NH}_3)_6]^{3+}$

(R) $[\text{NiCl}_4]^{2-}$

(S) $[\text{NiCN}]_4]^{2-}$

List-II

(1) sp^3d^2

(2) d^2sp^3

(3) sp^3

(4) dsp^2

Choose the correct answer from the options given below:

(1) $\text{P} \rightarrow 1$; $\text{Q} \rightarrow 2$; $\text{R} \rightarrow 3$; $\text{S} \rightarrow 4$

(2) $\text{P} \rightarrow 2$; $\text{Q} \rightarrow 1$; $\text{R} \rightarrow 4$; $\text{S} \rightarrow 3$

(3) $\text{P} \rightarrow 1$; $\text{Q} \rightarrow 2$; $\text{R} \rightarrow 4$; $\text{S} \rightarrow 3$

(4) $\text{P} \rightarrow 2$; $\text{Q} \rightarrow 1$; $\text{R} \rightarrow 3$; $\text{S} \rightarrow 4$

Ans. (1)

Sol. $[\text{CoF}_6]^{3-} \rightarrow 3\text{d}^6 \rightarrow \text{sp}^3\text{d}^2$

$[\text{Co}(\text{NH}_3)_6]^{3+} \rightarrow 3\text{d}^6 \rightarrow \text{d}^2\text{sp}^3$

$[\text{NiCl}_4]^{2-} \rightarrow 3\text{d}^8 \rightarrow \text{sp}^3$

$[\text{NiCN}]_4]^{2-} \rightarrow 3\text{d}^8 \rightarrow \text{dsp}^2$